M18 Glyceryl Trinitrate (GTN)

Scope of Practice	EMT
Classification	• Organic Nitrate / Vasodilator
Usual Presentation	• 0.4mg (per spray) spray bottle

Mechanism of Action	• GTN acts upon vascular smooth muscle to cause venous dilation (and arterial dilation to a lesser extent).
	• This results in:
	  • Reduced venous return (preload) this reduces ventricular filling and cardiac output which reduces myocardial oxygen demand
	  • Reduced peripheral resistance (afterload) through arterial dilation. This reduces myocardial oxygen demand through reducing the force the ventricles must overcome to eject blood.
	  • Dilation to coronary arteries which may increase myocardial blood flow.

Indications	• Cardiac Chest Pain • Pulmonary edema
Contraindications	• Known allergy to corn or corn products • SBP <90mmHg • Use of Phosphodiesterase inhibitors within the last 24 hours (e.g. Viagra) • Right ventricular infarct • Heart rate <50 or >150
Precautions	• Physiologically unstable patients / patients with signs of poor perfusion.
Side Effects	• Hypotension • Headache • Tachycardia • Lightheaded feeling
Adult Dose & Administration	**Chest Pain & Pulmonary Edema** • SL: 0.4mg(1x spray) every 3-5 minutes (max 1.2mg/ 3x sprays)
Pediatric Dose & Administration	• Not for pediatric administration
Use in Pregnancy and Lactation	• Safety has not been established. The likelihood of myocardial infarction or ischemia in pregnancy or lactation is very low, GTN should generally be withheld
Additional Information	• Before administering GTN SL spray, do not shake the bottle as this may lead to sub-therapeutic dosing. • Consider performing a Right Sided 12-Lead ECG to rule out right ventricular infarction.